

Md. Refath Islam

(He / Him / His)

Department of Mathematics,
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[Google Scholar](#)
[ResearchGate](#)

Research Interest

- Evolutionary game theory and population dynamics
- Replicator dynamics and Network Monte Carlo methods
- Nonlinear dynamics and bifurcation theory
- Multi-agent reinforcement learning
- Stochastic and symmetric dynamical systems
- Cyclic dominance and bistable/oscillatory transitions

Academic Profile

2023 – 06 to 2025 – 12	Master of Science (MSc), Mathematics Department of Mathematics, Bangladesh University of Engineering and Technology (BUET) Thesis: Influence of Cooperative Space and Vaccination Behavior on Epidemic Disease Dynamics under an Emergency CGPA: 3.75 / 4.00 (4.00 scale) – 18 credit hours theory, 18 credit hours thesis (85% marks)
2018 – 01 to 2022 – 10	Bachelor of Science (BS), Mathematics Kabi Nazrul Government College (affiliated with the University of Dhaka) CGPA: 3.32 / 4.00 (4.00 scale) – 129 credit hours completed (67% marks)

Experience

2023 – 11 to 2025 – 12	Graduate Teaching Assistant (GTA) Department of Mathematics, BUET, Dhaka, Bangladesh • Assessed assignments and class tests; invigilated class tests under faculty supervision.
2025 – 02 to 2025 – 06	Research Assistant (RA) Exploring Vaccine Hesitancy, Acceptance, and Behavioral Patterns Towards Human Papillomavirus (HPV) in Bangladesh Funded by the University Grants Commission (UGC), Bangladesh • Designed and administered a multi-city survey to assess HPV vaccine knowledge, hesitancy, acceptance, and behavioral intentions, identifying key socio-demographic and decision-making factors. • Developed and implemented an SVIR compartmental model incorporating gender-differentiated vaccination strategies, waning vaccine immunity, and reproduction-number analysis to evaluate HPV vaccination outcomes. • Applied evolutionary game theory, statistical analysis, and machine learning models (logistic regression and decision trees), integrating survey and simulation findings to generate policy-relevant insights for gender-targeted HPV vaccination.

Publications

2025 – 06	Assessing the Social Dilemmas in Eco-Evolutionary Dynamics of Cooperation, Cheating, and Resistance under Antibiotic Pressure Saleha Begum, Md. Refath Islam , K. M. Ariful Kabir Applied Mathematics and Computation (Elsevier) – doi.org/10.1016/j.amc.2025.129793
2025 – 07	Strategic Evolution under Varying Dilemma Strength and Social Efficiency Deficit in Asymmetric Dyadic Game Mst. Niger Sultana, Md. Refath Islam , K. M. Ariful Kabir Chaos, Solitons and Fractals (Elsevier) – doi.org/10.1016/j.chaos.2025.117594
2026 – 01	Unveiling Socio-Demographic Determinants of Low Birth Weight Using Machine Learning Techniques Mohammad Safi Uddin, Md. Refath Islam , K. M. Ariful Kabir PLOS Global Public Health – doi.org/10.1371/journal.pgph.0005745
2026 – 07	The Role of Punishment and Loners in Shaping Cooperation across Network Topologies Mehedi Hasan, Md. Refath Islam , K. M. Ariful Kabir Chaos, Solitons and Fractals (Elsevier) – doi.org/10.1016/j.chaos.2026.118922
2026 – 02	Q-Learning Driven Adaptive Decision Rules and Environmental Transformation in Three-Strategy Evolutionary Games Saleha Begum, Md. Refath Islam , K. M. Ariful Kabir Chaos, Solitons and Fractals (Elsevier) – doi.org/10.1016/j.chaos.2026.118052

Under Review

2024 – 12	Influence of Cooperative Space and Vaccination Behavior on Epidemic Disease Dynamics under an Emergency Md. Refath Islam , K. M. Ariful Kabir Submitted to Heliyon
2025 – 09	Trait-Mediated Evolutionary Dynamics in the Optional Prisoner's Dilemma (OPD) Game Md. Refath Islam , Mst. Niger Sultana, Fahimur Rahman Shuvo, K. M. Ariful Kabir Submitted to Royal Society Open Science

Conference Presentations

05–07 Dec 2024	Exploring Tumor Evolution and Resistance through a Three-Strategy Evolutionary Game Theory Model Md. Refath Islam , K. M. Ariful Kabir 3rd International Conference on Mathematical Analysis and Applications in Modeling (ICMAAM 2024)
05–06 Feb 2025	Analyzing Maternal and Demographic Factors Affecting Low Birth Weight (LBW) in Bangladesh Using Machine Learning Approach Md. Refath Islam , Mohammad Safi Uddin, K. M. Ariful Kabir A. F. Mujibur Rahman – Bangladesh Mathematical Society 3rd National Mathematics Conference 2024

Awards & Honors

2026 – 06	Research Excellence Award Forum 86 Excellence Ceremony 2026 Awarded as a research contributor for work on the Influence of Cooperative Space and Vaccination Behavior on Epidemic Disease Dynamics under an Emergency BUET, Dhaka, Bangladesh
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Certificate

2024 – 09	Introduction to Artificial Intelligence – Grade: A+ Bangladesh-Korea Information Access Center (BK-IAC), Department of Computer Science and Engineering, BUET
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Leadership & Event Management Experience

05–06 Feb 2025	Conference Organizer A. F. Mujibur Rahman – Bangladesh Mathematical Society 3rd National Mathematics Conference 2024, Department of Mathematics, BUET Coordinated logistics, managed registration, assisted with promotions, arranged catering services, facilitated technical support, and ensured smooth event execution.
16 Jan 2026	Organizer 16th National Undergraduate Mathematics Olympiad 2025, Department of Mathematics, BUET Supervised examination activities, assisted in result processing and award arrangements, and worked closely with the organizing committee to ensure smooth event execution.

Skills and Expertise

Software	Programming Languages
- Visual Studio Code - PyCharm - Microsoft Office - Mathematica	- Python - C++ - Fortran

References

MSc Supervisor	Dr. K. M. Ariful Kabir Associate Professor, Department of Mathematics Bangladesh University of Engineering and Technology (BUET), Dhaka 1000, Bangladesh Email: km_ariful@math.buet.ac.bd Phone: +880 1722 852818
MSc Course Teacher	Dr. Md. Manirul Alam Sarker Professor, Department of Mathematics Bangladesh University of Engineering and Technology (BUET), Dhaka 1000, Bangladesh Email: masarker@math.buet.ac.bd Phone: +880 1715 362127
Research Collaborator	Md. Nahid Hasan Assistant Professor, Department of Mathematics Bangladesh University of Engineering and Technology (BUET), Dhaka 1000, Bangladesh Email: nhasan@math.buet.ac.bd Phone: +880 1521 218927